CS19541 - COMPUTER NETWORKS LAB

EXPERIMENT NO: 10

AIM

a) Internetworking with routers in CISCO PACKET TRACER simulator.
b) Design and configure a simple internetwork using a router, and design/configure an internetwork using wireless router, DHCP server, and internet cloud.

REQUIREMENTS (HARDWARE & SOFTWARE)

- **Software:** Cisco Packet Tracer Simulator (v7.x or higher)
- **Hardware / Network Devices:** Router (Cisco 2811 / Generic), Wireless Router (WRT300N), Cable Modem, Internet Cloud (PT-Cloud), Server (Cisco.com), End Devices (Desktop PCs, Laptop), Copper Straight-Through Cables, and Coaxial Cables.

PROCEDURE

Part A: Simple Internetworking Using a Router (Router & 2 PCs)

- Place 1 Router (Router1) and 2 PCs (PC0 and PC1) on the logical workspace.
- Connect FastEthernet0 port of PC0 to FastEthernet0/0 of Router1 using a copper straight-through cable.
- Connect FastEthernet0 port of PC1 to FastEthernet0/1 of Router1 using a copper straight-through cable.
- Configure Router1 interfaces (FastEthernet0/0 with 192.168.10.1/24 and FastEthernet0/1 with 192.168.20.1/24) via CLI.
- Configure IP address, subnet mask, and default gateway on PC0 (Gateway: 192.168.10.1) and PC1 (Gateway: 192.168.20.1).
- Verify end-to-end connectivity by sending a PDU from PC0 to PC1.

Part B: Internetwork with Wireless Router, Cloud, and DHCP Server

- Build the logical topology by adding PC, Laptop, Wireless Router (WRT300N), Cable Modem, Internet Cloud, and Cisco.com Server.
- Cabling: Connect PC to Wireless Router LAN port (Ethernet 1) via straight-through cable. Connect Wireless Router Internet port to Cable Modem (Port 1). Connect Cable Modem (Port 0) to Cloud coaxial port via coaxial cable. Connect Cloud Ethernet port to Cisco.com server (FastEthernet0) via straight-through cable.
- Configure Wireless Router: Set SSID to 'HomeNetwork', enable DHCP server with range starting at 192.168.0.100, and set static DNS to 208.67.220.220.
- Configure Laptop: Replace Ethernet module with WPC300N wireless module in Physical tab, power on, and connect to 'HomeNetwork' SSID via PC Wireless utility.
- Configure Cloud: In Config > Cable, map From Port (Coaxial7) to To Port (Ethernet6), and set provider network to Cable under Ethernet6.

- Configure Server: Set static IP 208.67.220.220, enable DHCP service (pool starting at 208.67.220.1), and enable DNS service with an 'A' record pointing 'cisco.com' to 208.67.220.220.

PROGRAM / CLI CONFIGURATION COMMANDS

Commands executed on Router1 Command Line Interface (CLI):

```
Router> enable
Router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)# interface FastEthernet0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
Router(config-if)# exit
Router(config)# interface FastEthernet0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)# end
Router# write memory
```

IP CONFIGURATION

Table 1: Router Configuration Table (Part A)

Device Name	Interface	IP Address	Subnet Mask
Router1	FastEthernet0/0	192.168.10.1	255.255.255.0
Router1	FastEthernet0/1	192.168.20.1	255.255.255.0

Table 2: PC Configuration Table (Part A)

Device Name	Interface	IP Address	Subnet Mask	Default Gateway
PC 0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1
PC 1	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1

Table 3: Addressing Table for Wireless & Cloud Internetwork (Part B)

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC	FastEthernet0	DHCP (192.168.0.101)	255.255.255.0	192.168.0.1
Laptop	Wireless0	DHCP	255.255.255.0	192.168.0.1

Wireless Router	LAN	192.168.0.1	255.255.255.0	N/A
Wireless Router	Internet	DHCP	DHCP	DHCP
Cisco.com Server	FastEthernet0	208.67.220.220	255.255.255.0	208.67.220.1

VERIFICATION & TESTING

Network verification was validated through simulation and command line testing:

1. PDU Simulation Verification (Part A): A simple ICMP PDU was transmitted from PC0 to PC1 across Router1 interfaces. Successful transmission and acknowledgment reply packets were confirmed in the simulation panel.

2. DHCP Verification via Command Prompt (Part B): On the PC, issuing 'ipconfig /release' followed by 'ipconfig /renew' successfully acquired IP: 192.168.0.101, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.0.1, and DNS Server: 208.67.220.220.

3. End-to-End DNS & ICMP Ping Verification (Part B): From the PC command prompt, executing 'ping Cisco.com' resulted in 4 successful replies with 0% packet loss:

```
C:\> ping Cisco.com
Pinging 208.67.220.220 with 32 bytes of data:
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127
Reply from 208.67.220.220: bytes=32 time=10ms TTL=127
Reply from 208.67.220.220: bytes=32 time=2ms TTL=127
Reply from 208.67.220.220: bytes=32 time=1ms TTL=127

Ping statistics for 208.67.220.220:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 10ms, Average = 3ms
```

RESULT

The internetwork topologies using routers, wireless router, DHCP server, and internet cloud were successfully designed and configured using Cisco Packet Tracer simulator. Network connectivity, dynamic address allocation via DHCP, and domain name resolution were verified with 100% successful packet delivery.